



Nabi 2 tablet (Courtesy: www.nabitablet.com)

Nabi 2 is a tablet with an integrated learning system that helps parents of autistic children to gain better control over the time that the kids spend with technology. The tablet has features such as the Chore List and Treasure Box that integrate incentives into a child's playtime.

Nabi 2 uses a basic token system in which predetermined chores are set. The list can be changed over time as kids finish playing with the loaded activities. It also allows parents to use a timer that prompts children to hand the tablet over to a parent or sibling after a configured duration of time is over. This is very helpful in the case of autistic children, because once they get comfortable with a technology there is a constant fear of them getting addicted to it.

Another interesting product is NeuroSky's Focus Pocus—an interactive game controlled by the player's brainwaves. Priced less than \$100, the goal of this product is to help people with concentration problems, especially children with Attention Deficit Hyperactivity Disorder (ADHD), to improve their impulse control, attention span and memory. The device utilises EEG to read brain wave states, electrophysiological activity and metabolic rate—and converts changes in these into instructions for the play characters to follow. Hence the children have to concentrate in order to get the characters to do what they want.

However, it is important to note that great care has to be taken while developing tools for special children—as one problem could lead to another. At last year's Icare4Autism's Interna-

tional Autism Conference, Professor Tamar Weiss of the Haifa University expressed her concern that while kids are attracted to technology and it could help them discover new concepts, it comes with a huge risk of them becoming glued to the devices for long hours ignoring social interactions—which is indeed more important for such kids. She explained that the trick is to devise games or activities that can be played on accessible

touchscreen devices but involve more than one player. Her team, along with another at the University of Trento, is using Mitsubishi's touchscreen table, to devise several activities and puzzles that will draw autistic children out of their shell and improve their social skills by encouraging them to play in teams or with more than one player.

An indigenous iPad application for children with ASD

Last year, SAP Labs India unveiled an iPad application called Bol that helps autistic children to learn and communicate better. This simple, easy-to-use application was developed by the team based on their experiences in working with the Autism Society of India for over a year. The application is highly user-friendly and customisable to suit the needs of autistic children in India.

Jayananda Kotri, product manager, SAP Labs India, says, "Working with ASD-affected children has been a learning experience throughout. We learnt to respect their disabilities and appreciate the special skills they possess. Our first step was to explore the use of technology (primarily tablets like iPad) to assist them in learning through their formative years. We realised that the biggest challenge families of ASD-affected children face is to make them independent from their support so that they can lead a normal life. We realised soon that communication is a primary need and is surprisingly least addressed in the available applications. There are only a handful of apps available, which are also expensive

and localised to a Western audience. For example, the available apps had references to Western food items like burgers and kiwi fruit with an American voice-over, which gave an unfamiliar experience to the children. We conceptualised this communication app with a goal to make it available to all, free of cost, language, accent or region."

Bol, which can be downloaded for free from <https://itunes.apple.com/us/app/bol/id579955668>, integrates two key elements—auditory (record a voice-over description) and visual (take a picture). This allows autistic children to not just associate objects with sounds but also gain a comprehensive understanding of the materials being taught to them.

Kotri explains that the app uses pictures and voice for effective communication. The interface primarily displays a set of images in the form of a gallery/flash cards for learning or communicating. Pictures can be added into the app from a photo library or by taking a photograph using the built-in camera. Speech capability comes via voices that can be recorded to give the app a more human and personal touch or using the default text-to-speech (TTS) converter. The app stores everything into one file, which can also be exported via iTunes for backup. Future versions will allow sharing as well.

The app includes several small but significant features developed with a close understanding of the needs of ASD-affected children. For example, the app enlarges the image or item touched and also plays the recorded voice to enable them to communicate with others or for reinforced learning. If there are further items under that category, those are then displayed below that. This needs no special training. A child can explore everything on his own. The voice recording capability helps the app to connect to the child in a better manner giving a more personal feeling. The overall user experience is kept very simple and clutter-free to avoid distraction.

When asked about current work on this, Kotri says, "The Android version of the app is under development